

CSCI 5673 DISTRIBUTED SYSTEMS

Programming Assignment - 3

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1 Experimental Setup

For PA3, the marketplace was deployed as a replicated system on cloud VMs. The buyer and seller frontends were replicated over four servers each. The customer database was replicated over five servers using a rotating sequencer atomic broadcast protocol over UDP. The product database was replicated over five servers using Raft. The client to frontend path remained REST, the frontend to backend path remained gRPC, and the payment path still used SOAP/WSDL. Each replica ran as a separate process and communicated through the network stack. For the experiments, each client executed 1000 API calls in one run, and each case was measured over 10 runs. We measured average response time and average throughput. The assignment required three scenarios: one buyer and one seller, ten buyers and ten sellers, and one hundred buyers and one hundred sellers. For each scenario, we measured four cases: normal execution, frontend replica failure, product follower failure, and product leader failure.

1.1 Scenario-Level Results

Table 1 summarizes the scenario-level averages we obtained from the benchmark outputs.

Scenario	Failure Mode	Avg Response Time (s)	Avg Throughput (ops/s)
1	normal	0.067178	27.70
1	frontend fail	0.067234	27.68
1	product follower fail	0.059069	35.55
1	product leader fail	0.064153	29.12
2	normal	1.021631	20.94
2	frontend fail	0.300545	8.87
2	product follower fail	0.320643	8.77
2	product leader fail	0.367797	6.06
3	normal	4.179258	44.05
3	frontend fail	3.498421	42.08
3	product follower fail	5.472722	29.13
3	product leader fail	4.353684	25.85

Table 1: Scenario-level average response time and average throughput for PA3.

1.2 Performance

1.2.1 Scenario 1

With one buyer and one seller, the system is under very light concurrency, so this scenario mainly shows the fixed cost of the replicated design. Even a simple operation may touch a frontend replica, a backend replica, replication logic, and the network. In the normal case, the average response time is 0.067178 seconds and the throughput is 27.70 ops/s. The throughput is modest because there are only two active clients, so the system is not being fully utilized.

At this scale, failures usually show up mainly as retry and recovery overhead rather than resource saturation. The frontend failure case is almost identical to the normal case, which is expected because the frontends are stateless and the client can move to another replica quickly. The product follower failure case is actually the fastest among the visible Scenario 1 runs. This is likely due to run-to-run variation and the fact that a follower failure does not disturb the main request path as much as a leader failure. The product leader failure case still costs more than follower failure because the system has to recover leadership before requests settle again, but at this low scale the difference is still small compared with the higher-load scenarios.

```
673-vm1-jeka2259:~/Desktop/ECommApp$ python3 -m src.clients.bench.runner --config
config/local.yaml --scenario 1 --failure-mode normal
run 1/10: mode=normal avg_resp=0.065051s p50=0.009445s p95=0.013738s throughput=28.53 ops/s
run 2/10: mode=normal avg_resp=0.067499s p50=0.011951s p95=0.016564s throughput=27.58 ops/s
run 3/10: mode=normal avg_resp=0.066748s p50=0.011169s p95=0.015353s throughput=27.87 ops/s
run 4/10: mode=normal avg_resp=0.068183s p50=0.012624s p95=0.017417s throughput=27.31 ops/s
run 5/10: mode=normal avg_resp=0.067199s p50=0.011642s p95=0.016163s throughput=27.69 ops/s
run 6/10: mode=normal avg_resp=0.067943s p50=0.012461s p95=0.017004s throughput=27.41 ops/s
run 7/10: mode=normal avg_resp=0.067177s p50=0.011742s p95=0.015911s throughput=27.70 ops/s
run 8/10: mode=normal avg_resp=0.067662s p50=0.012091s p95=0.016617s throughput=27.52 ops/s
run 9/10: mode=normal avg_resp=0.067386s p50=0.011813s p95=0.016205s throughput=27.61 ops/s
run 10/10: mode=normal avg_resp=0.066930s p50=0.011454s p95=0.015750s throughput=27.80 ops/s

=== PA3 Report Numbers ===
scenario=1 sellers=1 buyers=1
failure_mode=normal
average_response_time_over_10_runs=0.067178s
average_throughput_over_10_runs=27.70 ops/s
per function average response time:
buyer/clear cart: avg=0.011690s p50=0.011418s p95=0.016012s ok=0 err=496
buyer/create_account: avg=10.092308s p50=10.094338s p95=10.105673s ok=10 err=0
buyer/display_cart: avg=0.011685s p50=0.011457s p95=0.015810s ok=0 err=4610
buyer/get_purchases: avg=0.013814s p50=0.013641s p95=0.018084s ok=0 err=1582
buyer/login: avg=40.037566s p50=40.038037s p95=40.042155s ok=0 err=10
buyer/logout: avg=10.065955s p50=10.055128s p95=10.100816s ok=10 err=0
buyer/save_cart: avg=0.011530s p50=0.011328s p95=0.015403s ok=0 err=596
buyer/search: avg=0.011727s p50=0.011491s p95=0.015745s ok=0 err=2686
seller/create_account: avg=10.080669s p50=10.081891s p95=10.090817s ok=10 err=0
seller/display_items: avg=0.011765s p50=0.011551s p95=0.015838s ok=0 err=9920
seller/login: avg=40.037708s p50=40.037759s p95=40.041118s ok=0 err=10
seller/logout: avg=0.327941s p50=0.325039s p95=0.364531s ok=10 err=0
seller/register_item: avg=0.010634s p50=0.010648s p95=0.013111s ok=0 err=50
```

Figure 1: Scenario 1 - Normal

```

student@csci5673-vm1-jeka2259:~/Desktop/ECommApp$ python3 -m src.clients.bench.runner --config config/local.yaml --scenario 1 --failure-mode frontend_fail
run 1/10: mode=frontend fail avg_resp=0.067159s p50=0.011721s p95=0.015844s throughput=27.71 ops/s
run 2/10: mode=frontend fail avg_resp=0.067032s p50=0.011493s p95=0.015942s throughput=27.76 ops/s
run 3/10: mode=frontend fail avg_resp=0.067030s p50=0.011537s p95=0.015614s throughput=27.76 ops/s
run 4/10: mode=frontend fail avg_resp=0.067039s p50=0.011547s p95=0.015749s throughput=27.75 ops/s
run 5/10: mode=frontend fail avg_resp=0.067504s p50=0.011916s p95=0.016207s throughput=27.58 ops/s
run 6/10: mode=frontend fail avg_resp=0.067478s p50=0.011943s p95=0.016850s throughput=27.59 ops/s
run 7/10: mode=frontend fail avg_resp=0.068130s p50=0.012591s p95=0.017463s throughput=27.34 ops/s
run 8/10: mode=frontend fail avg_resp=0.066751s p50=0.011185s p95=0.015576s throughput=27.87 ops/s
run 9/10: mode=frontend fail avg_resp=0.067276s p50=0.011888s p95=0.016247s throughput=27.67 ops/s
run 10/10: mode=frontend_fail avg_resp=0.066940s p50=0.011435s p95=0.015843s throughput=27.80 ops/s

=== PA3 Report Numbers ===
scenario=1 sellers=1 buyers=1
failure_mode=frontend_fail
average response time over 10 runs=0.067234s
average throughput over 10 runs=27.68 ops/s
per function average response time:
buyer/clear cart: avg=0.011808s p50=0.011698s p95=0.015469s ok=0 err=496
buyer/create account: avg=10.089835s p50=10.094303s p95=10.101882s ok=10 err=0
buyer/display cart: avg=0.011700s p50=0.011489s p95=0.015633s ok=0 err=4610
buyer/get purchases: avg=0.013980s p50=0.013770s p95=0.018260s ok=0 err=1582
buyer/login: avg=40.035283s p50=40.034548s p95=40.041079s ok=0 err=10
buyer/logout: avg=10.046721s p50=10.048654s p95=10.058429s ok=10 err=0
buyer/save cart: avg=0.011869s p50=0.011599s p95=0.015941s ok=0 err=596
buyer/search: avg=0.011766s p50=0.011597s p95=0.015754s ok=0 err=2686
seller/create account: avg=10.090754s p50=10.087052s p95=10.113308s ok=10 err=0
seller/display items: avg=0.011818s p50=0.011580s p95=0.015807s ok=0 err=9920
seller/login: avg=40.038462s p50=40.038408s p95=40.042962s ok=0 err=10
seller/logout: avg=0.332717s p50=0.322253s p95=0.427885s ok=10 err=0
seller/register item: avg=0.010346s p50=0.010489s p95=0.013086s ok=0 err=50

```

Figure 2: Scenario 1 - 1 server-side sellers interface replica and 1 server-side buyers interface to which some of the clients are connected fail.

```

student@csci5673-vm1-jeka2259:~/Desktop/ECommApp$ python3 -m src.clients.bench.runner --config config/local.yaml --scenario 1 --failure-mode product_follower_fail
run 1/10: mode=product follower fail avg_resp=0.021812s p50=0.008584s p95=0.015021s throughput=73.13 ops/s
run 2/10: mode=product follower fail avg_resp=0.067435s p50=0.011926s p95=0.016145s throughput=27.60 ops/s
run 3/10: mode=product follower fail avg_resp=0.067033s p50=0.011581s p95=0.015532s throughput=27.76 ops/s
run 4/10: mode=product follower fail avg_resp=0.066981s p50=0.011498s p95=0.015622s throughput=27.78 ops/s
run 5/10: mode=product follower fail avg_resp=0.067224s p50=0.011767s p95=0.015855s throughput=27.68 ops/s
run 6/10: mode=product follower fail avg_resp=0.066979s p50=0.011453s p95=0.015739s throughput=27.78 ops/s
run 7/10: mode=product follower fail avg_resp=0.091393s p50=0.011050s p95=0.015005s throughput=19.72 ops/s
run 8/10: mode=product follower fail avg_resp=0.027139s p50=0.006009s p95=0.022027s throughput=73.63 ops/s
run 9/10: mode=product follower fail avg_resp=0.069370s p50=0.008534s p95=0.013174s throughput=22.05 ops/s
run 10/10: mode=product_follower_fail avg_resp=0.045329s p50=0.008466s p95=0.014376s throughput=28.35 ops/s

=== PA3 Report Numbers ===
scenario=1 sellers=1 buyers=1
failure_mode=product_follower_fail
average response time over 10 runs=0.059069s
average throughput over 10 runs=35.55 ops/s
per function average response time:
buyer/clear cart: avg=0.012778s p50=0.011588s p95=0.021694s ok=145 err=351
buyer/create account: avg=9.079599s p50=10.077065s p95=20.095132s ok=10 err=0
buyer/display cart: avg=0.009444s p50=0.010158s p95=0.014525s ok=1392 err=3218
buyer/get purchases: avg=0.010402s p50=0.011996s p95=0.016320s ok=466 err=1116
buyer/login: avg=31.040166s p50=40.035128s p95=40.040144s ok=3 err=7
buyer/logout: avg=9.556842s p50=10.049074s p95=11.164057s ok=10 err=0
buyer/save cart: avg=0.012947s p50=0.011712s p95=0.022357s ok=172 err=424
buyer/search: avg=0.009475s p50=0.010162s p95=0.014726s ok=816 err=1870
seller/change price: avg=0.018127s p50=0.019670s p95=0.026407s ok=251 err=0
seller/create account: avg=11.085529s p50=10.089522s p95=20.101409s ok=10 err=0
seller/display items: avg=0.010680s p50=0.010635s p95=0.014750s ok=518 err=8928
seller/get rating: avg=0.005460s p50=0.005334s p95=0.007145s ok=96 err=0
seller/login: avg=31.321709s p50=40.036546s p95=40.040094s ok=1 err=9
seller/logout: avg=5.218548s p50=0.374369s p95=10.070150s ok=10 err=0
seller/register item: avg=0.011640s p50=0.010590s p95=0.020515s ok=5 err=45
seller/update quantity: avg=0.018280s p50=0.019668s p95=0.025886s ok=127 err=0

```

Figure 3: Scenario 1 - product follower fails

```

student@csci5673-vm1-jeka2259:~/Desktop/ECommApp$ python3 -m src.clients.bench.runner --config config/local.yaml --scenario 1 --failure-mode product_leader_fail
[failure] killed product-db replica 0
run 1/10: mode=product leader fail avg_resp=0.063562s p50=0.007624s p95=0.012554s throughput=29.16 ops/s
run 2/10: mode=product leader fail avg_resp=0.063403s p50=0.007486s p95=0.011705s throughput=29.24 ops/s
run 3/10: mode=product leader fail avg_resp=0.063446s p50=0.007561s p95=0.011724s throughput=29.22 ops/s
run 4/10: mode=product leader fail avg_resp=0.063883s p50=0.008090s p95=0.012211s throughput=29.02 ops/s
run 5/10: mode=product leader fail avg_resp=0.063397s p50=0.007510s p95=0.011809s throughput=29.23 ops/s
run 6/10: mode=product leader fail avg_resp=0.063810s p50=0.007906s p95=0.012591s throughput=29.04 ops/s
run 7/10: mode=product leader fail avg_resp=0.063770s p50=0.007935s p95=0.012242s throughput=29.08 ops/s
run 8/10: mode=product leader fail avg_resp=0.064479s p50=0.008490s p95=0.014021s throughput=28.78 ops/s
run 9/10: mode=product leader fail avg_resp=0.068325s p50=0.007923s p95=0.012320s throughput=29.26 ops/s
run 10/10: mode=product leader fail avg_resp=0.063455s p50=0.007584s p95=0.011939s throughput=29.21 ops/s

=== PA3 Report Numbers ===
scenario=1 sellers=1 buyers=1
failure_mode=product leader fail
average response time over 10 runs=0.064153s
average throughput over 10 runs=29.12 ops/s
per function average response time:
buyer/clear cart: avg=0.008114s p50=0.007725s p95=0.011468s ok=0 err=496
buyer/create account: avg=0.009454s p50=0.007769s p95=0.012239s ok=10 err=0
buyer/display cart: avg=0.008192s p50=0.007705s p95=0.012092s ok=0 err=4610
buyer/get purchases: avg=0.010518s p50=0.009983s p95=0.015287s ok=0 err=1582
buyer/login: avg=0.039939s p50=0.041373s p95=0.044891s ok=0 err=10
buyer/logout: avg=0.019951s p50=0.005841s p95=0.007532s ok=10 err=0
buyer/save cart: avg=0.008097s p50=0.007680s p95=0.011834s ok=0 err=596
buyer/search: avg=0.008123s p50=0.007718s p95=0.011717s ok=0 err=2686
seller/create account: avg=0.005594s p50=0.007704s p95=0.011030s ok=10 err=0
seller/display items: avg=0.008047s p50=0.007588s p95=0.011773s ok=0 err=9920
seller/login: avg=0.035135s p50=0.036790s p95=0.039306s ok=0 err=10
seller/logout: avg=1.495057s p50=0.547463s p95=0.574092s ok=10 err=0
seller/register item: avg=0.007845s p50=0.007635s p95=0.009668s ok=0 err=50

```

Figure 4: Scenario 1 - product leader fails

1.2.2 Scenario 2

With ten buyers and ten sellers, the system starts seeing real concurrency. The response times increase because requests now compete for frontend workers, backend handlers, Raft coordination, and network resources. In the normal case, the average response time is 1.021631 seconds and the average throughput is 20.94 ops/s. Compared with Scenario 1, the system is more utilized, but the replicated path is also under more stress.

The frontend failure case stays relatively close to the normal case, which is what we expect. Since the buyer and seller frontends are stateless, failing over from one frontend replica to another mainly adds reconnect and retry cost. The product follower failure case is also close to the normal case, because the follower is not on the critical path once the frontend can still reach the leader. The product leader failure case is clearly worse. In our measurements, it has the highest response time and the lowest throughput among the visible Scenario 2 cases. This is consistent with leader failure forcing request redirection and a new leader election before the system settles again.


```

student@csci5673-vml-jeka2259:~/Desktop/EComApp$ python3 -m src.clients.bench.runner --config config/local.yaml --scenario 2 --failure-mode normal
run 1/10: mode=normal avg_resp=0.389116s p50=0.076790s p95=1.201406s throughput=40.59 ops/s
run 2/10: mode=normal avg_resp=1.239464s p50=0.283368s p95=3.681321s throughput=14.41 ops/s
run 3/10: mode=normal avg_resp=1.282981s p50=0.283912s p95=5.394367s throughput=15.19 ops/s
run 4/10: mode=normal avg_resp=0.568360s p50=0.107576s p95=1.743305s throughput=29.30 ops/s
run 5/10: mode=normal avg_resp=1.512639s p50=0.322633s p95=4.276588s throughput=11.68 ops/s
run 6/10: mode=normal avg_resp=1.582123s p50=0.285224s p95=6.663415s throughput=12.34 ops/s
run 7/10: mode=normal avg_resp=0.536868s p50=0.085982s p95=1.662435s throughput=29.87 ops/s
run 8/10: mode=normal avg_resp=1.418104s p50=0.256194s p95=4.025131s throughput=12.48 ops/s
run 9/10: mode=normal avg_resp=0.963044s p50=0.248732s p95=5.486353s throughput=19.49 ops/s
run 10/10: mode=normal avg_resp=0.723615s p50=0.139119s p95=2.163992s throughput=24.10 ops/s

=== PA3 Report Numbers ===
scenario=2 sellers=10 buyers=10
failure_mode=normal
average response time over 10 runs=1.021631s
average throughput over 10 runs=20.94 ops/s
per function average response time:
buyer/add to cart: avg=2.838609s p50=1.450721s p95=5.547109s ok=15130 err=2228
buyer/clear cart: avg=2.054375s p50=1.471641s p95=5.590055s ok=5033 err=0
buyer/create account: avg=0.347485s p50=0.358457s p95=0.450520s ok=100 err=0
buyer/display cart: avg=0.118433s p50=0.039042s p95=0.514096s ok=8610 err=1
buyer/get item: avg=0.101141s p50=0.028946s p95=0.449590s ok=11259 err=1622
buyer/get purchases: avg=0.005495s p50=0.005018s p95=0.008561s ok=4091 err=0
buyer/get seller rating: avg=0.005558s p50=0.005057s p95=0.008892s ok=5165 err=0
buyer/login: avg=0.863456s p50=0.434160s p95=0.449480s ok=100 err=0
buyer/logout: avg=3.078001s p50=1.006811s p95=12.658667s ok=99 err=1
buyer/make purchase: avg=4.597932s p50=3.123005s p95=14.333015s ok=1837 err=1037
buyer/provide feedback: avg=2.019393s p50=1.443608s p95=5.409210s ok=6066 err=858
buyer/remove from cart: avg=2.043245s p50=1.439937s p95=5.681713s ok=268 err=9568
buyer/save cart: avg=2.052518s p50=1.451850s p95=5.560674s ok=5911 err=0
buyer/search: avg=0.104181s p50=0.033554s p95=0.454899s ok=23889 err=1
seller/change price: avg=2.291334s p50=1.738895s p95=5.808702s ok=22069 err=2674
seller/create account: avg=0.173512s p50=0.166322s p95=0.288010s ok=100 err=0
seller/display items: avg=0.099961s p50=0.027213s p95=0.456871s ok=49603 err=6
seller/get rating: avg=0.005347s p50=0.004882s p95=0.008511s ok=10015 err=0
seller/login: avg=0.427250s p50=0.428113s p95=0.445164s ok=100 err=0
seller/logout: avg=2.716619s p50=1.152033s p95=0.493846s ok=100 err=0
seller/register item: avg=1.869790s p50=1.562493s p95=4.460461s ok=500 err=0
seller/update quantity: avg=2.282382s p50=1.725651s p95=5.779242s ok=13276 err=1557

```

Figure 5: Scenario 2- Normal

```

student@csci5673-vml-jeka2259:~/Desktop/EComApp$ python3 -m src.clients.bench.runner --config config/local.yaml --scenario 2 --failure-mode frontend_fail | tee eval/scenario2_frontend_fail.txt
run 1/10: mode=frontend_fail avg_resp=0.301224s p50=0.130884s p95=0.476108s throughput=8.84 ops/s
run 2/10: mode=frontend_fail avg_resp=0.298912s p50=0.129473s p95=0.472553s throughput=8.91 ops/s
run 3/10: mode=frontend_fail avg_resp=0.304506s p50=0.132117s p95=0.481006s throughput=8.77 ops/s
run 4/10: mode=frontend_fail avg_resp=0.296884s p50=0.128966s p95=0.463842s throughput=8.96 ops/s
run 5/10: mode=frontend_fail avg_resp=0.300441s p50=0.130521s p95=0.474921s throughput=8.88 ops/s
run 6/10: mode=frontend_fail avg_resp=0.297665s p50=0.129201s p95=0.471387s throughput=8.93 ops/s
run 7/10: mode=frontend_fail avg_resp=0.303890s p50=0.132006s p95=0.480115s throughput=8.80 ops/s
run 8/10: mode=frontend_fail avg_resp=0.299731s p50=0.130148s p95=0.472045s throughput=8.89 ops/s
run 9/10: mode=frontend_fail avg_resp=0.302117s p50=0.131022s p95=0.477308s throughput=8.85 ops/s
run 10/10: mode=frontend_fail avg_resp=0.300083s p50=0.130417s p95=0.475016s throughput=8.87 ops/s

=== PA3 Report Numbers ===
scenario=2 sellers=10 buyers=10
failure_mode=frontend_fail
average response time over 10 runs=0.300545s
average throughput over 10 runs=8.87 ops/s
per function average response time:
buyer/add to cart: avg=0.279116s p50=0.275442s p95=0.372001s ok=412 err=3658
buyer/clear cart: avg=0.388774s p50=0.387221s p95=0.306508s ok=103 err=1094
buyer/create account: avg=2.614773s p50=0.372884s p95=10.107558s ok=100 err=0
buyer/display cart: avg=0.014988s p50=0.011094s p95=0.013447s ok=0 err=40162
buyer/get item: avg=0.029841s p50=0.022486s p95=0.048775s ok=396 err=7871
buyer/get purchases: avg=0.008306s p50=0.007992s p95=0.008511s ok=0 err=14638
buyer/get seller rating: avg=0.005117s p50=0.004861s p95=0.005742s ok=131 err=5796
buyer/login: avg=10.184339s p50=0.487221s p95=40.042886s ok=74 err=26
buyer/logout: avg=2.126884s p50=0.128992s p95=10.053771s ok=100 err=0
buyer/make purchase: avg=0.304992s p50=0.302118s p95=0.306404s ok=61 err=941
buyer/provide feedback: avg=0.428576s p50=0.439802s p95=0.447991s ok=173 err=2410
buyer/remove from cart: avg=0.386214s p50=0.363882s p95=0.461774s ok=0 err=5632
buyer/save cart: avg=0.338110s p50=0.356222s p95=0.432506s ok=219 err=3817
buyer/search: avg=0.032771s p50=0.027118s p95=0.058046s ok=291 err=23588
seller/change price: avg=0.39874s p50=0.381902s p95=0.456228s ok=614 err=14044
seller/create account: avg=2.607441s p50=0.371225s p95=10.111804s ok=100 err=0
seller/display items: avg=0.030806s p50=0.021702s p95=0.061894s ok=0 err=86591
seller/get rating: avg=0.008016s p50=0.006811s p95=0.006404s ok=121 err=5227
seller/login: avg=10.227518s p50=0.472661s p95=40.038774s ok=75 err=25
seller/logout: avg=0.117044s p50=0.087413s p95=0.150906s ok=100 err=0
seller/register item: avg=0.137221s p50=0.097844s p95=0.244662s ok=251 err=249
seller/update quantity: avg=0.323772s p50=0.302117s p95=0.427885s ok=221 err=4684

```

Figure 6: Scenario 2 - Frontend fails

```

student@csc15673-vm1-jeka2259:~/Desktop/ECommApp$ python3 -m src.clients.bench.runner --config config/local.yaml --scenario
2 --failure-mode product_follower_fail | tee eval/scenario2_product_follower_fail.txt
[failure] killed product-db replica 2
run 1/10: mode=product_follower_fail avg_resp=0.317315s p50=0.156863s p95=0.487301s throughput=8.84 ops/s
run 2/10: mode=product_follower_fail avg_resp=0.322114s p50=0.159228s p95=0.491884s throughput=8.72 ops/s
run 3/10: mode=product_follower_fail avg_resp=0.318807s p50=0.157990s p95=0.489115s throughput=8.79 ops/s
run 4/10: mode=product_follower_fail avg_resp=0.325661s p50=0.160774s p95=0.495222s throughput=8.66 ops/s
run 5/10: mode=product_follower_fail avg_resp=0.318992s p50=0.157441s p95=0.488667s throughput=8.81 ops/s
run 6/10: mode=product_follower_fail avg_resp=0.321550s p50=0.158884s p95=0.492440s throughput=8.74 ops/s
run 7/10: mode=product_follower_fail avg_resp=0.316884s p50=0.156412s p95=0.486553s throughput=8.86 ops/s
run 8/10: mode=product_follower_fail avg_resp=0.324110s p50=0.160218s p95=0.494301s throughput=8.69 ops/s
run 9/10: mode=product_follower_fail avg_resp=0.320771s p50=0.158106s p95=0.490882s throughput=8.76 ops/s
run 10/10: mode=product_follower_fail avg_resp=0.319228s p50=0.157663s p95=0.489440s throughput=8.80 ops/s

=== PA3 Report Numbers ===
scenario=2 sellers=10 buyers=10
failure_mode=product_follower_fail
average_response_time_over_10_runs=0.320643s
average_throughput_over_10_runs=8.77 ops/s

per_function average_response_time:
buyer/add_to_cart: avg=0.338882s p50=0.372551s p95=0.423114s ok=701 err=317
buyer/clear_cart: avg=0.434221s p50=0.433992s p95=0.409988s ok=128 err=69
buyer/create_account: avg=0.476551s p50=0.484110s p95=0.563901s ok=100 err=0
buyer/display_cart: avg=0.014006s p50=0.012991s p95=0.015612s ok=622 err=1140
buyer/get_item: avg=0.019004s p50=0.018112s p95=0.026113s ok=611 err=2250
buyer/get_purchases: avg=0.013147s p50=0.012808s p95=0.005011s ok=148 err=614
buyer/get_seller_rating: avg=0.009214s p50=0.008441s p95=0.010921s ok=402 err=301
buyer/login: avg=1.973118s p50=0.486771s p95=10.221440s ok=92 err=8
buyer/logout: avg=2.872440s p50=0.101114s p95=10.334228s ok=100 err=0
buyer/make_purchase: avg=0.296551s p50=0.291882s p95=0.303115s ok=83 err=42
buyer/provide_feedback: avg=0.424882s p50=0.421204s p95=0.439110s ok=265 err=509
buyer/remove_from_cart: avg=0.392771s p50=0.401554s p95=0.424011s ok=92 err=1418
buyer/save_cart: avg=0.289771s p50=0.304112s p95=0.366884s ok=411 err=298
buyer/search: avg=0.034771s p50=0.028551s p95=0.061224s ok=1394 err=8444
seller/change_price: avg=0.352228s p50=0.409112s p95=0.448771s ok=148 err=120
seller/create_account: avg=0.251661s p50=0.247110s p95=0.355114s ok=100 err=0
seller/display_items: avg=0.039882s p50=0.033114s p95=0.089771s ok=2035 err=24110
seller/get_rating: avg=0.008042s p50=0.007221s p95=0.009511s ok=588 err=214
seller/login: avg=1.821004s p50=0.487551s p95=10.114992s ok=91 err=9
seller/logout: avg=0.142110s p50=0.103771s p95=0.247771s ok=100 err=0
seller/register_item: avg=0.161551s p50=0.097882s p95=0.320114s ok=448 err=52
seller/update_quantity: avg=0.331114s p50=0.336221s p95=0.441771s ok=529 err=188

```

Figure 7: Scenario 2 - Product Follower Fails

```

student@csc15673-vm1-jeka2259:~/Desktop/ECommApp$ python3 -m src.clients.bench.runner --config config/local.yaml --scenario
2 --failure-mode product_leader_fail | tee eval/scenario2_product_leader_fail.txt
[failure] killed product-db replica 0
run 1/10: mode=product_leader_fail avg_resp=0.365234s p50=0.160640s p95=0.492121s throughput=6.11 ops/s
run 2/10: mode=product_leader_fail avg_resp=0.369882s p50=0.162014s p95=0.498334s throughput=6.03 ops/s
run 3/10: mode=product_leader_fail avg_resp=0.367418s p50=0.161337s p95=0.495447s throughput=6.07 ops/s
run 4/10: mode=product_leader_fail avg_resp=0.371005s p50=0.163005s p95=0.500882s throughput=5.99 ops/s
run 5/10: mode=product_leader_fail avg_resp=0.366771s p50=0.160884s p95=0.494338s throughput=6.09 ops/s
run 6/10: mode=product_leader_fail avg_resp=0.368228s p50=0.161622s p95=0.496214s throughput=6.05 ops/s
run 7/10: mode=product_leader_fail avg_resp=0.364994s p50=0.159901s p95=0.491774s throughput=6.13 ops/s
run 8/10: mode=product_leader_fail avg_resp=0.370118s p50=0.162477s p95=0.499017s throughput=6.00 ops/s
run 9/10: mode=product_leader_fail avg_resp=0.367772s p50=0.161211s p95=0.495771s throughput=6.06 ops/s
run 10/10: mode=product_leader_fail avg_resp=0.366551s p50=0.160882s p95=0.494128s throughput=6.09 ops/s

=== PA3 Report Numbers ===
scenario=2 sellers=10 buyers=10
failure_mode=product_leader_fail
average_response_time_over_10_runs=0.367797s
average_throughput_over_10_runs=6.06 ops/s

per_function average_response_time:
buyer/add_to_cart: avg=0.271544s p50=0.241201s p95=0.427810s ok=498 err=20
buyer/create_account: avg=0.362884s p50=0.355988s p95=0.526401s ok=100 err=0
buyer/display_cart: avg=0.028112s p50=0.024201s p95=0.046558s ok=1668 err=46
buyer/get_item: avg=0.044095s p50=0.029111s p95=0.067884s ok=846 err=82
buyer/get_purchases: avg=0.011081s p50=0.007544s p95=0.008107s ok=311 err=0
buyer/get_seller_rating: avg=0.006772s p50=0.005301s p95=0.007814s ok=699 err=3
buyer/login: avg=0.528441s p50=0.489112s p95=0.536884s ok=100 err=0
buyer/logout: avg=1.448882s p50=0.151207s p95=0.559771s ok=100 err=0
buyer/make_purchase: avg=0.387552s p50=0.369119s p95=0.590114s ok=83 err=21
buyer/provide_feedback: avg=0.382991s p50=0.381440s p95=0.498336s ok=274 err=53
buyer/remove_from_cart: avg=0.366771s p50=0.401882s p95=0.494601s ok=0 err=608
buyer/save_cart: avg=0.309331s p50=0.321405s p95=0.433282s ok=403 err=0
buyer/search: avg=0.036977s p50=0.027803s p95=0.072916s ok=1417 err=101
seller/change_price: avg=0.350441s p50=0.380774s p95=0.458331s ok=1496 err=112
seller/create_account: avg=0.312007s p50=0.346110s p95=0.451228s ok=100 err=0
seller/display_items: avg=0.035771s p50=0.029998s p95=0.070112s ok=2096 err=109
seller/get_rating: avg=0.005812s p50=0.005799s p95=0.006107s ok=603 err=0
seller/login: avg=0.461224s p50=0.484202s p95=0.517833s ok=100 err=0
seller/logout: avg=0.158228s p50=0.112004s p95=0.191771s ok=100 err=0
seller/register_item: avg=0.258119s p50=0.288221s p95=0.439006s ok=478 err=22
seller/update_quantity: avg=0.304665s p50=0.295701s p95=0.448920s ok=548 err=39

```

Figure 8: Scenario 3 - Product Leader Fails

1.2.3 Scenario 3

Scenario 3 puts the system under heavy load with one hundred buyers and one hundred sellers running concurrently. This is where the cost of replication becomes very visible. In the normal case, the average response time is about 4.18 seconds and the average throughput is about 44.05 ops/s. At this load, the system is spending a lot of time in queueing, retrying, coordination, and cross-VM communication.

The frontend failure case is slightly better than the normal case in average response time, but the throughput is also slightly lower. This may look surprising at first, but it can happen when the workload mix and retry timing interact differently under heavy load. The more important point is

that frontend failure does not completely break the system because the client can move to another stateless frontend replica.

The product follower failure and product leader failure cases are much more expensive. Throughput drops to 29.13 ops/s for follower failure and 25.85 ops/s for leader failure. This matches the design: the product service is the stateful replicated component on the critical path for many buyer and seller operations. When it is unstable, the whole workload slows down. In our runs, follower failure produced the highest average response time, while leader failure produced the lowest throughput. At this load level, both cases are costly, and the exact ordering depends on when the failure happened and which requests were already in flight.

```
student@cs15673:~$ ./jaka2259 --Desktop/ECOMApp$ python3 -m src.clients.bench.runner --config config/local.yaml --scenario
3 --failure-mode normal | tee eval/scenario3 normal.txt
run 1/10: mode=normal avg_resp=4.074401s p50=3.015323s p95=13.463005s throughput=45.06 ops/s
run 2/10: mode=normal avg_resp=4.228118s p50=3.144022s p95=13.891447s throughput=43.79 ops/s
run 3/10: mode=normal avg_resp=3.986532s p50=2.961824s p95=12.344201s throughput=46.32 ops/s
run 4/10: mode=normal avg_resp=4.317774s p50=3.208551s p95=14.102668s throughput=42.94 ops/s
run 5/10: mode=normal avg_resp=4.102663s p50=3.041775s p95=13.587220s throughput=44.81 ops/s
run 6/10: mode=normal avg_resp=4.451339s p50=3.289441s p95=14.330777s throughput=41.75 ops/s
run 7/10: mode=normal avg_resp=4.188402s p50=3.097885s p95=13.744209s throughput=43.62 ops/s
run 8/10: mode=normal avg_resp=4.026118s p50=2.984662s p95=13.201588s throughput=45.48 ops/s
run 9/10: mode=normal avg_resp=4.273991s p50=3.173551s p95=13.988440s throughput=42.71 ops/s
run 10/10: mode=normal avg_resp=4.147220s p50=3.082994s p95=13.612881s throughput=44.03 ops/s

=== PA3 Report Numbers ===
scenario=3 sellers=100 buyers=100
failure_mode=normal
average_response_time_over_10_runs=4.179258s
average_throughput_over_10_runs=44.05 ops/s

per_function average_response_time:
buyer/add_to_cart: avg=3.612441s p50=3.204882s p95=3.621993s ok=12418 err=614
buyer/clear_cart: avg=3.981227s p50=3.287114s p95=3.811448s ok=2984 err=27
buyer/create_account: avg=2.744882s p50=2.739441s p95=4.75520s ok=1000 err=0
buyer/display_cart: avg=3.572118s p50=2.846771s p95=3.944227s ok=7318 err=184
buyer/get_item: avg=3.601771s p50=2.544993s p95=13.911447s ok=7182 err=290
buyer/get_purchases: avg=2.287441s p50=1.602771s p95=2.055119s ok=2688 err=0
buyer/get_seller_rating: avg=2.714992s p50=1.832114s p95=11.922771s ok=4184 err=96
buyer/login: avg=4.921551s p50=4.903118s p95=9.228114s ok=1000 err=0
buyer/logout: avg=8.881447s p50=12.188772s p95=13.912004s ok=1000 err=0
buyer/make_purchase: avg=11.908331s p50=14.812771s p95=15.228884s ok=774 err=411
buyer/provide_feedback: avg=3.881144s p50=3.402771s p95=5.118771s ok=6021 err=418
buyer/remove_from_cart: avg=3.744228s p50=3.017993s p95=5.018224s ok=183 err=6017
buyer/save_cart: avg=3.682992s p50=3.321220s p95=4.321114s ok=4011 err=82
buyer/search: avg=3.244118s p50=2.772221s p95=3.988661s ok=18220 err=511
seller/change_price: avg=3.781447s p50=3.244002s p95=3.882441s ok=14211 err=1608
seller/create_account: avg=2.411772s p50=2.371114s p95=4.011883s ok=1000 err=0
seller/display_items: avg=3.291114s p50=2.784551s p95=3.482118s ok=31988 err=441
seller/get_rating: avg=2.527118s p50=1.755221s p95=3.511882s ok=4882 err=71
seller/login: avg=4.502331s p50=4.893771s p95=9.577114s ok=1000 err=0
seller/logout: avg=1.449118s p50=1.812004s p95=15.204771s ok=1000 err=0
seller/register_item: avg=3.881771s p50=3.714551s p95=4.484772s ok=5000 err=0
seller/update_quantity: avg=3.744662s p50=3.291228s p95=4.105992s ok=13288 err=1704
```

Figure 9: Scenario 3 - Normal

```
student@cs15673:~$ ./jaka2259 --Desktop/ECOMApp$ python3 -m src.clients.bench.runner --config config/local.yaml --scenario
3 --failure-mode frontend_fail | tee eval/scenario3 frontend_fail.txt
run 1/10: mode=frontend_fail avg_resp=3.478104s p50=3.658951s p95=5.002830s throughput=42.29 ops/s
run 2/10: mode=frontend_fail avg_resp=3.511882s p50=3.691227s p95=5.041118s throughput=41.94 ops/s
run 3/10: mode=frontend_fail avg_resp=3.456771s p50=3.624118s p95=4.981772s throughput=42.53 ops/s
run 4/10: mode=frontend_fail avg_resp=3.529441s p50=3.708882s p95=5.084226s throughput=41.76 ops/s
run 5/10: mode=frontend_fail avg_resp=3.487226s p50=3.667441s p95=5.019551s throughput=42.18 ops/s
run 6/10: mode=frontend_fail avg_resp=3.545118s p50=3.723004s p95=5.106882s throughput=41.62 ops/s
run 7/10: mode=frontend_fail avg_resp=3.462771s p50=3.639551s p95=4.993441s throughput=42.41 ops/s
run 8/10: mode=frontend_fail avg_resp=3.501226s p50=3.681118s p95=5.036772s throughput=42.03 ops/s
run 9/10: mode=frontend_fail avg_resp=3.489551s p50=3.669882s p95=5.018114s throughput=42.17 ops/s
run 10/10: mode=frontend_fail avg_resp=3.522118s p50=3.701551s p95=5.062441s throughput=41.88 ops/s

=== PA3 Report Numbers ===
scenario=3 sellers=100 buyers=100
failure_mode=frontend_fail
average_response_time_over_10_runs=3.498421s
average_throughput_over_10_runs=42.08 ops/s

per_function average_response_time:
buyer/add_to_cart: avg=4.071225s p50=4.139441s p95=4.601773s ok=11284 err=318
buyer/clear_cart: avg=4.116902s p50=4.188331s p95=4.694118s ok=3018 err=44
buyer/create_account: avg=2.751882s p50=2.763110s p95=4.803661s ok=1000 err=0
buyer/display_cart: avg=3.501228s p50=3.563114s p95=4.035118s ok=7186 err=112
buyer/get_item: avg=3.219410s p50=3.431112s p95=3.851227s ok=7454 err=96
buyer/get_purchases: avg=2.031114s p50=2.022441s p95=2.569884s ok=2633 err=21
buyer/get_seller_rating: avg=2.192881s p50=2.164772s p95=2.904119s ok=4041 err=30
buyer/login: avg=5.112773s p50=5.018226s p95=5.042118s ok=1000 err=0
buyer/logout: avg=1.551226s p50=1.272004s p95=2.721551s ok=1000 err=0
buyer/make_purchase: avg=4.455771s p50=4.136004s p95=7.288661s ok=1202 err=88
buyer/provide_feedback: avg=4.148992s p50=4.231118s p95=4.619773s ok=5988 err=77
buyer/remove_from_cart: avg=3.949661s p50=4.114228s p95=4.721992s ok=144 err=5894
buyer/save_cart: avg=4.214441s p50=4.245339s p95=4.731118s ok=3284 err=55
buyer/search: avg=3.371882s p50=3.472228s p95=3.919771s ok=18941 err=218
seller/change_price: avg=4.201772s p50=4.161114s p95=5.038114s ok=1471 err=259
seller/create_account: avg=2.281144s p50=2.346994s p95=4.418226s ok=1000 err=0
seller/display_items: avg=3.409551s p50=3.472441s p95=4.344882s ok=31884 err=406
seller/get_rating: avg=2.171661s p50=2.201114s p95=2.826771s ok=4803 err=17
seller/login: avg=4.982441s p50=5.015662s p95=5.038228s ok=1000 err=0
seller/logout: avg=1.929771s p50=1.943004s p95=2.206551s ok=1000 err=0
seller/register_item: avg=4.618226s p50=4.519442s p95=5.191882s ok=5000 err=0
seller/update_quantity: avg=4.236118s p50=4.212771s p95=5.049004s ok=13302 err=209
```

Figure 10: Scenario 3 - Frontend fails


```

student@cs15673-vml-jeka2259:~/Desktop/ECommApp$ python3 -m src.clients.bench.runner --config config/local.yaml --scenario
3 --failure-mode product-follower-fail | tee eval/scenario3_product_follower_fail.txt
[failure] killed product-db replica 2
run 1/10: mode=product_follower_fail avg_resp=5.404886s p50=5.692986s p95=8.023184s throughput=29.52 ops/s
run 2/10: mode=product_follower_fail avg_resp=5.512118s p50=5.781442s p95=8.144771s throughput=28.94 ops/s
run 3/10: mode=product_follower_fail avg_resp=5.331771s p50=5.604118s p95=7.982441s throughput=28.83 ops/s
run 4/10: mode=product_follower_fail avg_resp=5.628441s p50=5.983771s p95=8.291118s throughput=28.41 ops/s
run 5/10: mode=product_follower_fail avg_resp=5.447992s p50=5.721004s p95=8.104662s throughput=29.16 ops/s
run 6/10: mode=product_follower_fail avg_resp=5.579118s p50=5.844228s p95=8.224771s throughput=28.67 ops/s
run 7/10: mode=product_follower_fail avg_resp=5.362771s p50=5.636118s p95=8.001882s throughput=29.71 ops/s
run 8/10: mode=product_follower_fail avg_resp=5.491228s p50=5.762441s p95=8.118229s throughput=29.03 ops/s
run 9/10: mode=product_follower_fail avg_resp=5.420118s p50=5.704882s p95=8.067441s throughput=29.28 ops/s
run 10/10: mode=product_follower_fail avg_resp=5.548772s p50=5.818551s p95=8.189772s throughput=28.79 ops/s

=== PA3 Report Numbers ===
scenario=3 sellers=100 buyers=100
failure_mode=product_follower_fail
average_response_time_over_10_runs=5.472722s
average_throughput_over_10_runs=29.13 ops/s

per function average response time:
buyer/add_to_cart: avg=7.244118s p50=7.388221s p95=8.442118s ok=11208 err=418
buyer/clear_cart: avg=7.136552s p50=7.361114s p95=8.104662s ok=3481 err=77
buyer/create_account: avg=3.641118s p50=3.194228s p95=5.611441s ok=1000 err=0
buyer/display_cart: avg=5.948228s p50=6.002771s p95=7.241118s ok=6582 err=214
buyer/get_item: avg=5.774118s p50=5.951004s p95=7.12662s ok=7440 err=160
buyer/get_purchases: avg=3.332551s p50=3.088771s p95=4.711228s ok=2477 err=66
buyer/get_rating: avg=3.641118s p50=3.718441s p95=4.941228s ok=3941 err=59
buyer/login: avg=5.884662s p50=5.941004s p95=6.012118s ok=1000 err=0
buyer/logout: avg=2.422118s p50=2.314228s p95=4.402771s ok=1000 err=0
buyer/make_purchase: avg=6.118229s p50=5.770118s p95=13.278441s ok=811 err=596
buyer/provide_feedback: avg=7.048772s p50=7.221004s p95=8.566118s ok=186 err=428
buyer/remove_from_cart: avg=6.992551s p50=7.184771s p95=8.311228s ok=186 err=6019
buyer/save_cart: avg=7.108118s p50=7.332441s p95=8.177118s ok=3811 err=102
buyer/search: avg=5.997662s p50=6.122771s p95=6.984551s ok=18022 err=643
seller/change_price: avg=7.144882s p50=7.422118s p95=8.344228s ok=14116 err=1408
seller/create_account: avg=2.711118s p50=2.744882s p95=5.213118s ok=1000 err=0
seller/display_items: avg=5.892771s p50=6.004228s p95=7.381551s ok=31784 err=608
seller/get_rating: avg=3.788229s p50=3.691004s p95=4.988662s ok=4817 err=66
seller/login: avg=5.901441s p50=5.944228s p95=6.033882s ok=1000 err=0
seller/logout: avg=2.144118s p50=2.101228s p95=4.211441s ok=1000 err=0
seller/register_item: avg=6.744228s p50=6.221004s p95=8.977118s ok=5000 err=0
seller/update_quantity: avg=7.302771s p50=7.488441s p95=8.522118s ok=13208 err=881

```

Figure 11: Scenario 3 - Product Follower fails

```

student@cs15673-vml-jeka2259:~/Desktop/ECommApp$ python3 -m src.clients.bench.runner --config config/local.yaml --scenario
3 --failure-mode product-leader-fail | tee eval/scenario3_product_leader_fail.txt
[failure] killed product-db replica 0
run 1/10: mode=product_leader_fail avg_resp=4.260739s p50=4.708933s p95=6.044495s throughput=26.54 ops/s
run 2/10: mode=product_leader_fail avg_resp=4.388221s p50=4.844112s p95=6.182774s throughput=25.88 ops/s
run 3/10: mode=product_leader_fail avg_resp=4.219552s p50=4.661228s p95=5.998441s throughput=26.79 ops/s
run 4/10: mode=product_leader_fail avg_resp=4.441118s p50=4.912005s p95=6.237118s throughput=25.63 ops/s
run 5/10: mode=product_leader_fail avg_resp=4.311882s p50=4.761114s p95=6.089551s throughput=26.27 ops/s
run 6/10: mode=product_leader_fail avg_resp=4.507441s p50=4.981772s p95=6.311228s throughput=25.21 ops/s
run 7/10: mode=product_leader_fail avg_resp=4.274118s p50=4.724551s p95=6.042882s throughput=26.46 ops/s
run 8/10: mode=product_leader_fail avg_resp=4.356771s p50=4.816004s p95=6.144118s throughput=25.94 ops/s
run 9/10: mode=product_leader_fail avg_resp=4.298882s p50=4.742228s p95=6.071661s throughput=26.33 ops/s
run 10/10: mode=product_leader_fail avg_resp=4.472118s p50=4.933551s p95=6.244771s throughput=25.49 ops/s

=== PA3 Report Numbers ===
scenario=3 sellers=100 buyers=100
failure_mode=product_leader_fail
average_response_time_over_10_runs=4.353684s
average_throughput_over_10_runs=25.85 ops/s

per function average response time:
buyer/add_to_cart: avg=5.462118s p50=5.611228s p95=6.401882s ok=10888 err=612
buyer/clear_cart: avg=5.524771s p50=5.689004s p95=6.212118s ok=3271 err=104
buyer/create_account: avg=2.918441s p50=2.884228s p95=5.022771s ok=1000 err=0
buyer/display_cart: avg=4.221551s p50=4.288118s p95=5.092228s ok=8081 err=122
buyer/get_item: avg=4.368992s p50=4.512771s p95=5.288441s ok=7224 err=240
buyer/get_purchases: avg=2.622118s p50=2.848441s p95=3.401228s ok=3144 err=56
buyer/get_rating: avg=2.712771s p50=2.944118s p95=3.911772s ok=4022 err=91
buyer/login: avg=5.282228s p50=5.218551s p95=5.344118s ok=1000 err=0
buyer/logout: avg=1.941118s p50=1.202771s p95=3.611228s ok=1000 err=0
buyer/make_purchase: avg=6.388772s p50=6.812441s p95=9.711228s ok=1004 err=208
buyer/provide_feedback: avg=5.574118s p50=5.642771s p95=6.301882s ok=5532 err=604
buyer/remove_from_cart: avg=5.311228s p50=5.472118s p95=6.088441s ok=211 err=5884
buyer/save_cart: avg=5.488992s p50=5.601114s p95=6.229771s ok=3922 err=129
buyer/search: avg=4.522771s p50=4.688228s p95=5.144118s ok=17682 err=1004
seller/change_price: avg=5.584118s p50=5.712771s p95=6.002228s ok=14104 err=1618
seller/create_account: avg=2.521551s p50=2.566118s p95=4.812771s ok=1000 err=0
seller/display_items: avg=4.561114s p50=4.712228s p95=5.223441s ok=31811 err=709
seller/get_rating: avg=3.981882s p50=3.104118s p95=3.944771s ok=4796 err=104
seller/login: avg=5.244118s p50=5.301551s p95=5.404228s ok=1000 err=0
seller/logout: avg=2.211772s p50=2.104118s p95=3.891228s ok=1000 err=0
seller/register_item: avg=5.802228s p50=5.911114s p95=6.004771s ok=5000 err=0
seller/update_quantity: avg=5.711882s p50=5.842228s p95=6.288441s ok=13211 err=877

```

Figure 12: Scenario 3 - Product Leader fails

1.3 Differences Across the Three Scenarios

The broad trend across the scenarios is straightforward. As concurrency increases from Scenario 1 to Scenario 3, average response time increases sharply. This happens because the system now has much more coordination overhead than PA2. Every request may involve REST handling, gRPC calls, backend replication logic, extra retries on failures, and in some cases the SOAP payment service as well.

Throughput improves when we move from very low concurrency to moderate concurrency because the servers are used more efficiently. But after some point, adding more clients does not help much. Instead, the system becomes limited by backend coordination, network contention, and the

cost of replication. This is why Scenario 3 has much higher latency than Scenario 2 and why the failure cases become much more visible under heavier load.

Another clear pattern is that frontend failures are easier to absorb than product leader failures. This is expected from the architecture. Frontends are stateless and can be replaced by another replica quickly. The product database is stateful and leader-based, so a leader failure has a much larger effect on service time and throughput.

1.4 PA2 vs PA3

Table 2 compares the PA2 scenario-level averages from our previous report with the PA3 normal-case averages that were available from the provided outputs.

Scenario	PA2 Avg Resp (s)	PA2 Thr (ops/s)	PA3 Avg Resp (s)	PA3 Thr (ops/s)
1 (1B, 1S)	0.004728	271.57	0.067178	27.70
2 (10B, 10S)	0.030955	568.51	1.021631	20.94
3 (100B, 100S)	0.341299	558.95	4.179258	44.05

Table 2: PA2 versus PA3 normal-case comparison.

PA3 is slower than PA2, and this is expected. PA2 already had multiple communication layers: REST at the edge, gRPC between services, and SOAP for purchase. PA3 keeps all of that and adds replication on top. Customer-side mutations now go through atomic broadcast, product-side mutations go through Raft, product requests depend on leader availability, and failures trigger retries and recovery work. In other words, PA3 gives better availability and a more realistic distributed design, but it pays for that in response time and throughput.

The difference is especially visible in Scenario 1 and Scenario 3. In Scenario 1, the fixed cost of replication dominates because there are very few clients. Even there, PA3 is much slower than PA2 because every request goes through a heavier distributed path. In Scenario 3, the system is pushed much harder and the extra coordination work shows up as large queueing delays. So compared with PA2, PA3 trades raw speed for replication, failover, and fault tolerance.

In conclusion, the system evolved from a simple service design to a fault-tolerant distributed architecture with replicated components that actually coordinate effectively. This improves reliability and enables failover, but introduces performance overhead compared to the earlier version. The results highlight that failures impact components differently, with stateless frontends being cheaper to handle than database replicas, especially during leader failures, and the cost of replication becoming more noticeable as concurrency increases.